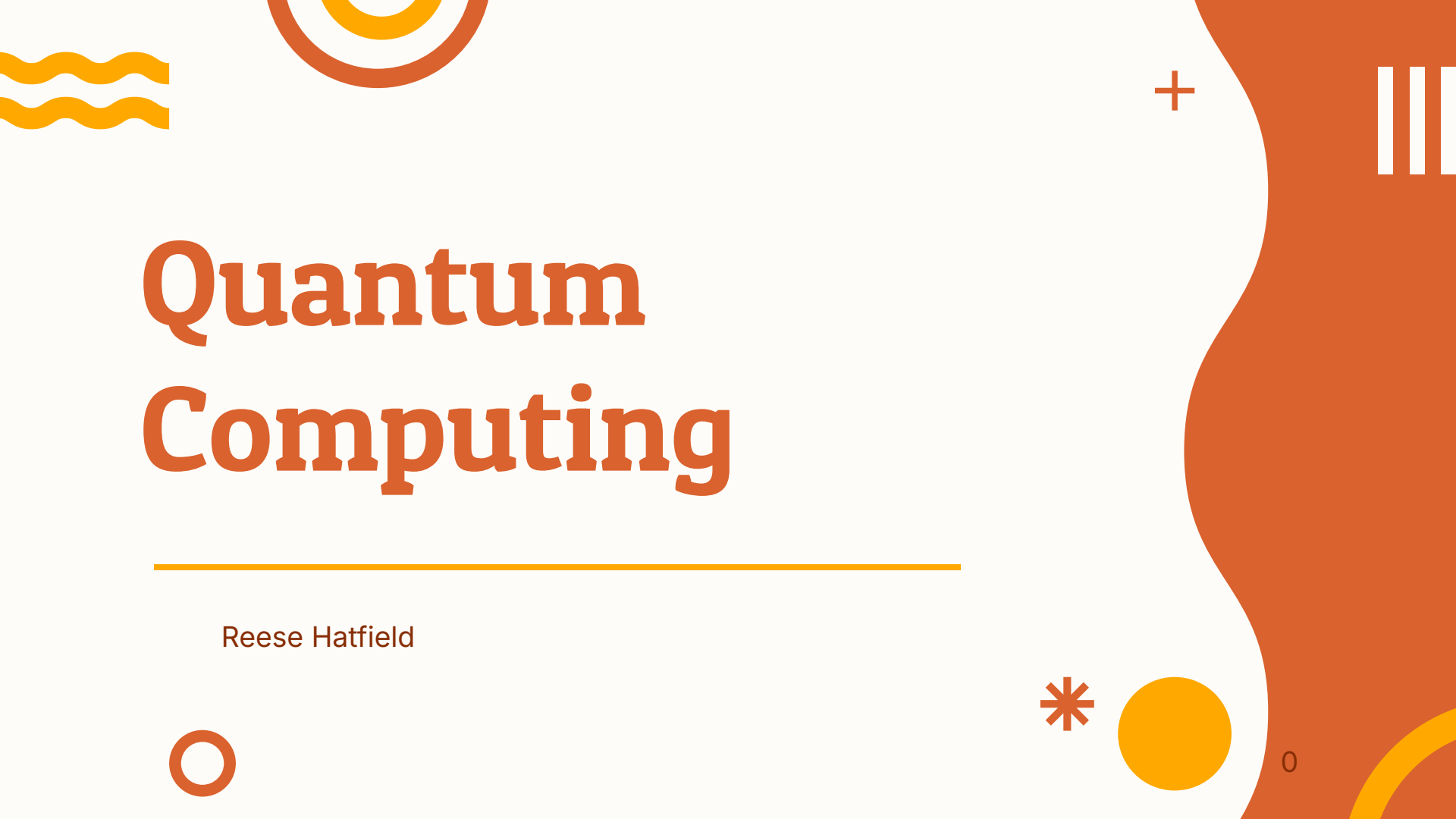
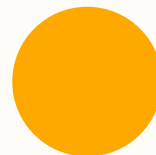




Quantum Computing

Reese Hatfield



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Quantum Computing

- In the beginning
- There were bits
- Purely mathematical
- You can do math on them





Quantum Computing

- At the start of the class
- We read a Claude Shannon paper
- Boolean Algebra
- Fundamental idea of computer science





Quantum Computing

- Shannon bridged a gap
- Used boolean algebra
- Analyzied a physical device
- This is the basis of how we *designed* computers





Quantum Computing

- Physical switching relays
- Turned them into bits
- Into pure math

- Abstractions
- Root of computer science





Quantum Computing



1 or 0





Quantum Computing

- What Shannon analyzed was, in a way, an arbitrary decision
- It just tended to work out nicely
- What if we changed WHAT a bit abstracted upon?





Quantum Computing

- Most things $\Rightarrow$ Similar results
- Analog Signals
- Continuous values

- Similar, but hardly any more useful
 - Outside or particular areas
- Let's think back





Quantum Computing

- Math, Physics, Comp. Sci
- Were at one point, the same
- What are some other things in physics you could abstract a bit over?





Quantum Computing

- Do they have different properties?
- At a quantum mechanical level?
- Maybe?





Quantum Computing

- Do they have different properties?
- At a quantum mechanical level?
- Maybe!
- Electron Spin?
- Photon Polarization?
- Superconducting Circuits?





Quantum Computing

- Enter a new type of bit
- Arises from these abstractions
- **Qubit**
- Quantum Bit
- Has different properties than a classical bit





Quantum Computing

- Qubits can be 1 or 0
 - Like a classical bit
- Or could *also* be a combination of both simultaneously
- This feels strange?
 - Remember, it arises naturally out of what we abstracted on





Quantum Computing

- Representing both at the same time
- A **superposition of both**





Quantum Computing

- Representing both at the same time
- A **superposition of both**
- Is that even a bit?
- What do we mean by a bit anyway?





Quantum Computing

- Its still a bit
- When a qubit is measure
 - Observed
- It “collapses” into a 1 or a 0
- All qubits will have a definite state upon observation





Quantum Computing

- But that fact that can encode multiple bits at the same time
- Is why quantum computing is so powerful
- Algorithmic Speedups
- If we can encode the problem correctly





Quantum Computing

- Superpositions allows us to represent more states at the same time
- String multiple qubits together
 - Just as you would with bits
- To solve more complex problems
- Encode complex ideas





Quantum Computing

- We need perform operations upon qubits
- We need gates, but at the qubit level
- Enter:
 - **Quantum Gates**
- Not as simple as AND, OR, NOT





Quantum Computing

- Quantum Gates manipulate probabilities of state collapse*
- Qubits are often represented via a state vector
- Quantum Gates manipulate the direction of that vector



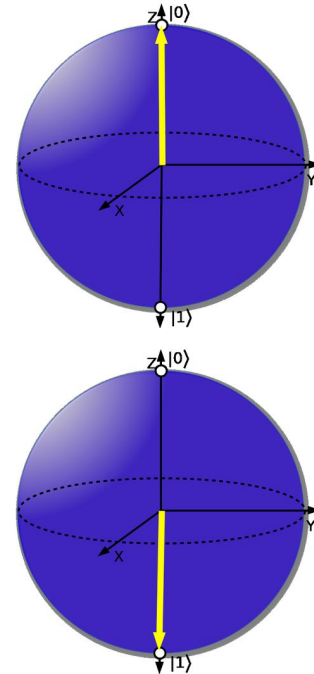
Quantum Computing

- Pauli-X Gate
- Effectively a quantum NOT gate
- Flips NORTH to SOUTH

Pauli X

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

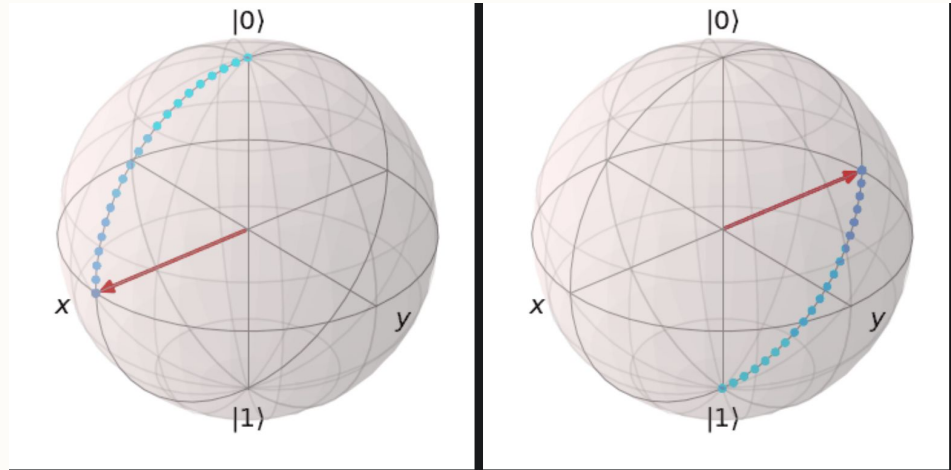
$$X |0\rangle = |1\rangle$$





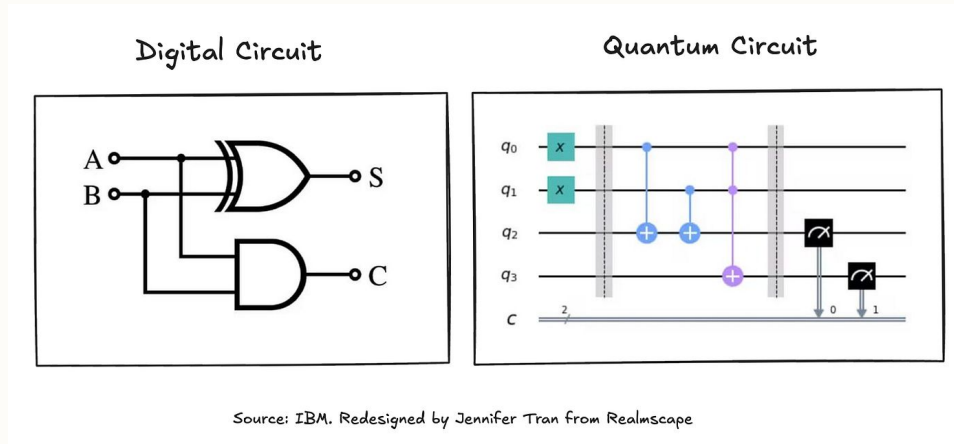
Quantum Computing

- Hadamard Gate
- Takes a definite bit
- Turns that 0 or 1 into a 50/50 superposition of both



Quantum Computing

- Put a bunch of gates together to build quantum circuits





Quantum Computing

- Quantum Circuits effectively generate a probability distribution of the output of qubits
- Upon measurement, circuit “collapses” into a single sequence of output qubits
- Random sample, but not uniform (see visual)





Quantum Computing

- Algorithms can work differently than algorithms on classical bits
- Shor's Algorithm
 - Factoring Large Numbers
- Grover's Algorithms
 - Quantum Search
- Operate faster than classical algorithms





Quantum Computing

- *Not all problems will get a speedup from quantum computing*
- Algorithm has to be able to leverage superposition in order to get a speed increase like this
- Why is this not mainstream yet?





Quantum Computing

- Qubits, in reality, are really expensive
- Scale very poorly
- 1990s- 2010s
 - Best quantum computers only had <20 qubits
- 2026 $\Rightarrow$ ~ 6000 qubits





Quantum Computing

